

Problem

A certain electronic circuit is connected to a 110-V ac line. The root-mean-square value of the current drawn is 2 A, with a phase angle of 55° .

(a) Find the true power drawn by the circuit.

(b) Calculate the apparent power.

Step-by-step solution

Step 1 of 2

$$\begin{aligned} \text{(A) } S &= (110)(2 \angle 55^\circ) = 220 \angle 55^\circ \\ P &= S \cos \theta = 220 \cos(55^\circ) \\ &= 126.2 \text{ W} . \end{aligned}$$

Step 2 of 2

$$\text{(B) } S = |S| = 220 \text{ VA} .$$